

Assignment1: CancerMortality

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HINT: Retain all available numeric variables except those that are not useful in explaining the target.

Load the dataset

```
rm(list=ls())
train <- read.csv("train.csv")
test <- read.csv("test.csv")
```

Data Preparation

Univariate Descriptive Analysis (Variable Analysis)

For each variable:

- Original numeric variables corresponding to qualitative concepts have to be converted to factors

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- Original numeric variables corresponding to real quantitative concepts are kept as numeric but additional factors should also be created

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- Exploratory Data Analysis for each variable (numeric summary and graphic support).

#Include Code

Data Quality Report

Per variable, count:

- Number of missing values
- Number of errors (including inconsistencies)
- Number of outliers -Rank variables according the sum of missing values (and errors)

#Include Code

Per individuals, count:

- Number of missing values
- Number of errors
- Number of outliers
- Identify individuals considered as multivariant outliers.

#Include Code

Create a variable adding the total number of missing values, outliers and errors. Describe these variables, to which other variables exist higher associations.

- Compute the correlation with all other variables. Rank these variables according the correlation
- Compute for every group of individuals the mean of missing/outliers/error values. Rank those groups according the computed mean.

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Imputation

- Numeric Variables
- Factors

#Include Code

Profiling

- Target : Death rate

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